



Determination of a Transition Plane in a Gas Atomization Simulation

4WC00, Bachelor Final Project

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Mechanical Engineering, Power and Flow

An aerial photograph of the TU/e campus in Eindhoven, Netherlands, taken at sunset. The image shows a large, modern glass-fronted building with a grid-like facade, surrounded by greenery and other campus buildings. The sky is a deep orange-red, and the city lights are visible in the background.

Eindhoven, August 27, 2026

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List of symbols

Symbol	Description	Unit
Flow and thermodynamic variables		
ρ	Density	kg m^{-3}
ρ_g	Gas density	kg m^{-3}
$\mathbf{u}$	Velocity vector	m s^{-1}
u_z	Axial droplet velocity	m s^{-1}
U_{rel}	Relative velocity between liquid and gas	m s^{-1}
p	Pressure	Pa
T	Temperature	K
μ	Dynamic viscosity	Pa s
τ	Viscous stress tensor	Pa
$\mathbf{g}$	Gravitational acceleration	m s^{-2}
Interface and surface-tension quantities		
α	Liquid volume fraction	–
$\alpha_{\text{threshold}}$	Threshold liquid volume fraction	–
F_{int}	Interfacial force	Pa
σ	Surface tension coefficient	N m^{-1}
κ	Interface curvature	m^{-1}
$\mathbf{n}$	Droplet interface normal vector	–
Droplet and spray quantities		
D	Droplet diameter	m
R_{min}	Minimum distance from droplet COM to interface	m
R_{max}	Maximum distance from droplet COM to interface	m
L_b	Breakup length	m
d_z	Axial distance from nozzle	m
We	Weber number	–
ψ	Shape criterion	–
VOF cloud extraction quantities		
V_i	Mesh cell volume	m^3
V_j	Liquid structure volume	m^3
$\mathbf{x}_i$	Position vector of cell i	m
$\mathbf{x}_{\text{COM}}$	Liquid structure center of mass	m
$\mathbf{U}_{\text{drop}}$	Droplet velocity	m s^{-1}
$\mathbf{U}_{\text{cont}}$	Surrounding gas velocity	m s^{-1}

Table 0.1: List of symbols

1 | Introduction

In today's rapidly developing manufacturing sector, Additive manufacturing (AM) remains a critical staple of the industry. Additive manufacturing is a manufacturing process that fabricates parts through the layer-by-layer addition of material [1]. Metal additive manufacturing (MAM) refers to this layer-by-layer manufacturing approach done using metal specifically. There are distinct techniques of MAM: powder bed fusion (PBF), directed energy deposition (DED), binder jetting, and numerous others, each with their own advantages and disadvantages. For example, while PBF offers superior accuracy and surface finish, DED may be utilized to deposit metal directly onto damaged parts to repair them. Each MAM technique has its own place in the broader manufacturing context.

A necessary feedstock to certain types of MAM, specifically PBF, binder jetting, and powder-fed DED, is metal powder. This metal powder allows for the manufacturing of complex metal parts with high precision, and must have certain properties:

- The metal powder must be fine.
- The metal powder must have high sphericity.
- The metal powder must be uniform, i.e. of consistent shape and size.

These manufacturing techniques require good flowability while maintaining minimal manufacturing defects. Hence, highly fine particles, with high sphericity i.e. near-spherical particle shape, and a consistent particle size distribution, are required properties for these systems. These properties affect how the powder spreads, packs, melts, and solidifies during the manufacturing process. Poor powder quality can lead to defects such as porosity, lack of fusion, uneven layers, and reduced mechanical performance in the final part.

Gas atomization is one of the most widely used methods of producing metal powder at an industrial scale [2]. In this process, a stream of molten metal is broken up using high-pressure gas jets, typically using inert gases. The molten metal stream gradually breaks up and separates into small droplets, which rapidly cool and solidify to form highly spherical powder particles. This complete process is shown in **Figure 1.1**. Gas atomization has several advantages: high production rates that make it suitable for large-scale manufacturing, production of fine and high sphericity powders, and control over the particle size distribution, which make it ideal for powder metallurgy and additive manufacturing [2].

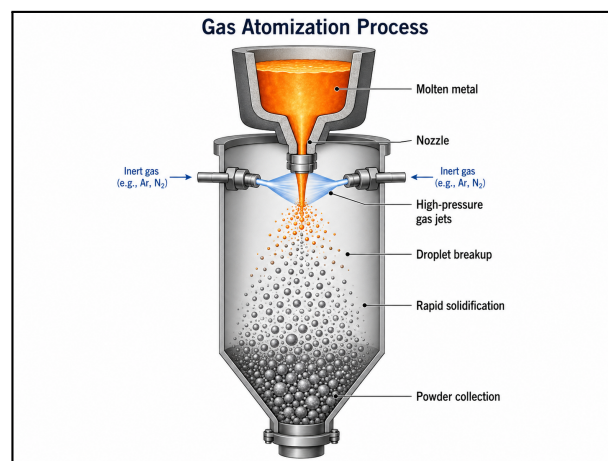


Figure 1.1: The gas atomization process. The image is AI-generated using ChatGPT [3].

There are alternative methods of producing fine metal powder as well. For instance: water atomization, centrifugal atomization, plasma atomization, ultrasonic atomization, and mechanical milling to name a few. These methods have their own place, water atomization and mechanical milling are cheap and efficient, however they produce highly irregular particles that are unsuited for precise application. Ultrasonic and plasma atomization produce fine, spherical and uniform particles, however these are expensive processes with low throughput, hence are less suited to high volume production.

Gas atomization is therefore a critical preceding step to MAM. MAM requires reliable powder feedstock, which gas atomization produces with the required shape, size, and quality. Thereby, understanding the gas atomization process helps explain how powder properties are controlled and why they directly influence the performance of additively manufactured components.

1.1 | The Gas Atomization Process

As stated, gas atomization refers to the process by which molten metal is broken up and solidified into fine metal particles. This process involves two key phases: the primary atomization phase, and the secondary atomization phase.

The primary atomization refers to the initial breakup of the continuous liquid metal stream. The gas-metal interface surrounding the liquid metal core is rapidly and violently destabilized due to momentum transfer by the surrounding high-velocity gas, causing the formation of metal ligaments and sheets that eventually detach into relatively large liquid metal droplets. This phase occurs closest to the nozzle [4].

The secondary atomization occurs following the primary atomization. The large liquid metal droplets break up into smaller droplets due to the high-velocity gas, as the disruptive aerodynamic forces of the gas overcome the restoring surface tension within the droplets [5]. The droplets deform into smaller droplets that undergo rapid cooling and solidification, which result in the production of fine and uniform particles throughout.

In real-life gas atomization, the primary and secondary atomization are part of a single continuous process. There is no exact point in space where a transition is observed from primary to secondary breakup. However, in computational simulations, a clear distinction between these phases is relevant since different numerical methods are utilized. In the simulation devised by Thuy [6] utilized in this report, primary atomization is modelled via a resolved Volume-of-Fluid (VOF) method, while the secondary atomization is modelled via Lagrangian Particle Tracking (LPT). This methodology is devised from the assumption that at the end of the primary atomization, liquid metal particles meet the required criteria for key properties, primarily droplet diameter (D), and sphericity (ψ), to ensure that these particles are sufficiently fine and spherical to be represented as point particles in the secondary atomization simulation.

To couple the two distinct numerical models, a transition plane is introduced to demarcate the primary and secondary atomization. The transition plane refers to a coupling to ensure that simulation data and results from the primary atomization may be transferred for use as the initial conditions for the secondary atomization. This is shown in Figure 1.2. The location of the transition plane must therefore satisfy two competing requirements. Firstly, it must be positioned sufficiently far downstream for the droplets to be sufficiently fine, spherical, and uniform. Secondly, it must remain as close to the nozzle as possible to avoid extending the simulation unnecessarily, thereby ensuring that the model is computationally efficient throughout. A well-defined transition plane ensures a balance between accuracy and computational efficiency, and is therefore highly critical in implementing effectively in a gas atomization simulation.

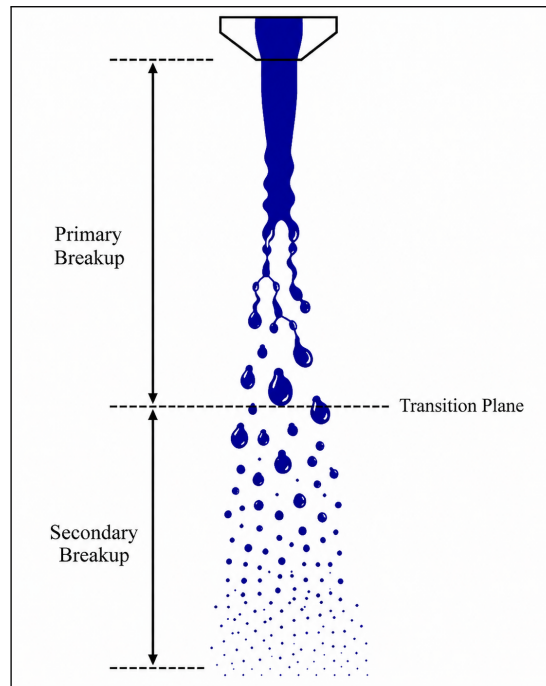


Figure 1.2: Schematic representation of primary and secondary breakup in gas atomization. The image is AI-generated using ChatGPT [3], adapted from [7].

The goal of this report is to make a recommendation for this transition plane utilizing relevant data from the primary atomization simulation, as well as validate this recommendation and draw a conclusion on its validity. From this, the research questions pertaining to this report may be ascertained.

1.2 | Research Questions

The report explores the following research questions:

RQ1. How do droplet properties evolve relative to axial distance?

RQ2. How may trends in droplet properties relative to axial distance be used to determine an optimal transition plane?

RQ3. Which additional parameters may be extracted to support the location of the transition plane?

2 | Theory

This section provides detail into the computational methodology utilized by OpenFOAM, which is the CFD software framework within which the simulations are run.

2.1 | Finite Volume Discretization and Mesh Refinement

The governing equations stated in [subsection 2.3](#) are continuous differential equations, which cannot be solved analytically in their continuous form. In OpenFOAM, this data is first discretized using the finite-volume method, by which the simulation domain is divided into a finite number of control volumes. i.e. cells. The governing equations are solved over each distinct cell and computed numerically.

The accuracy of the computational solution depends on the cell sizes utilized in the discretization, referred collectively to as the mesh. A finer mesh yields a higher simulation accuracy at the cost of higher computational expense. To improve the accuracy of data at specific regions of interest, in particular the cells located at the gas-melt interface, an adaptive mesh refinement (AMR) is utilized. This increases the mesh resolution locally at the gas-melt interface, which allows for small waves, ligaments and droplets to be represented more accurately without requiring the entire computational domain to use the finest mesh resolution [6].

2.2 | VOF Method

The Volume-of-Fluid (VOF) method is used to represent the gas-liquid interface during primary breakup. Both the gas and liquid metal phases are solved on the same Eulerian computational mesh. Within each computational cell in the mesh, a liquid volume fraction α is assigned as follows:

$$\alpha = \begin{cases} 1, & \text{liquid,} \\ 0, & \text{gas,} \\ 0 < \alpha < 1, & \text{gas-liquid interface.} \end{cases} \quad (2.1)$$

The time and spatial evolution of the liquid volume fraction field determines the location of each phase, which allows for large interface deformations, ligament formation and droplet detachment to be identified at a relatively low computational cost.

2.3 | Governing Equations

As stated, the liquid metal and gas in the simulation domain are modelled as continuous fluids on an Eulerian mesh. This states that the simulation domain is divided into fixed computational cells, and the fluid properties are calculated separately for each unique cell. Each cell stores relevant physical properties at a given time-step:

- Velocity $\mathbf{u}$
- Pressure p
- Density ρ
- Temperature T
- Liquid volume fraction α

OpenFOAM solves for the corresponding governing equations at each time-step. The evolution over time of the compressible flow is governed by the Navier-Stokes equations, which comprise the continuity equation, the momentum equation, and the energy equation.

The continuity equation is given by:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.2)$$

The conservation of momentum is given by:

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p_{\text{rgh}} - \nabla \cdot \boldsymbol{\tau} - \mathbf{g} \cdot h \nabla \rho + \mathbf{F}_{\text{int}} \quad (2.3)$$

where p_{rgh} is the hydrostatically corrected pressure, $\boldsymbol{\tau}$ is the viscous stress tensor, and $\mathbf{F}_{\text{int}}$ refers to the forces acting at the gas-liquid interface.

The thermal behaviour is given by the energy equation:

$$\frac{\partial(\rho T)}{\partial t} + \nabla \cdot (\rho \mathbf{u} T) = \nabla^2(aT) - \left(\frac{\alpha_n}{C_{v,1}} + \frac{1 - \alpha_n}{C_{v,2}} \right) \left(\frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) + \nabla \cdot (\mathbf{u} p) \right) \quad (2.4)$$

where $\boldsymbol{\tau}$ represents the viscous stress tensor.

Since the flow is compressible, the governing equations are solved with an equation of state (EoS), which provides the necessary relationship between pressure, density, and temperature required. The atomizing gas is modelled as an ideal gas, hence density is given as:

$$\rho_g = \frac{p}{RT} \quad (2.5)$$

where ρ_g is gas density, and R is the specific gas constant.

In addition to the pressure, viscous, and gravitational forces that appear in the momentum equation, the gas-liquid interface is also subject to surface tension.

Surface tension refers to the tendency of liquid surfaces to shrink to the lowest surface area due to cohesive internal forces [8]. In the case of droplets undergoing atomization, surface tension acts to resist deformation and fragmentation caused by the aerodynamic forces of high-velocity gas, and must be overcome to cause droplet breakup. The expression for surface tension may be expressed as:

$$\mathbf{F}_{\text{int}} = \sigma \kappa \mathbf{n} \quad (2.6)$$

where σ is the surface tension coefficient, κ is the local interface curvature, and $\mathbf{n}$ is the unit normal vector to the interface. The balance of force between aerodynamic forces and surface tension plays a key role in the deformation and breakup of the liquid phase.

2.4 | PISO Pressure-Velocity Coupling

The PISO algorithm is used to solve the compressible Navier-Stokes equations. At each time step, the momentum equation (Equation 2.3) is first solved to obtain a predicted velocity. This velocity does not necessarily satisfy mass conservation exactly, hence the PISO algorithm solves for a pressure-correction equation and subsequently corrects the pressure and velocity fields. The specifics of these equations are beyond the scope of this report.

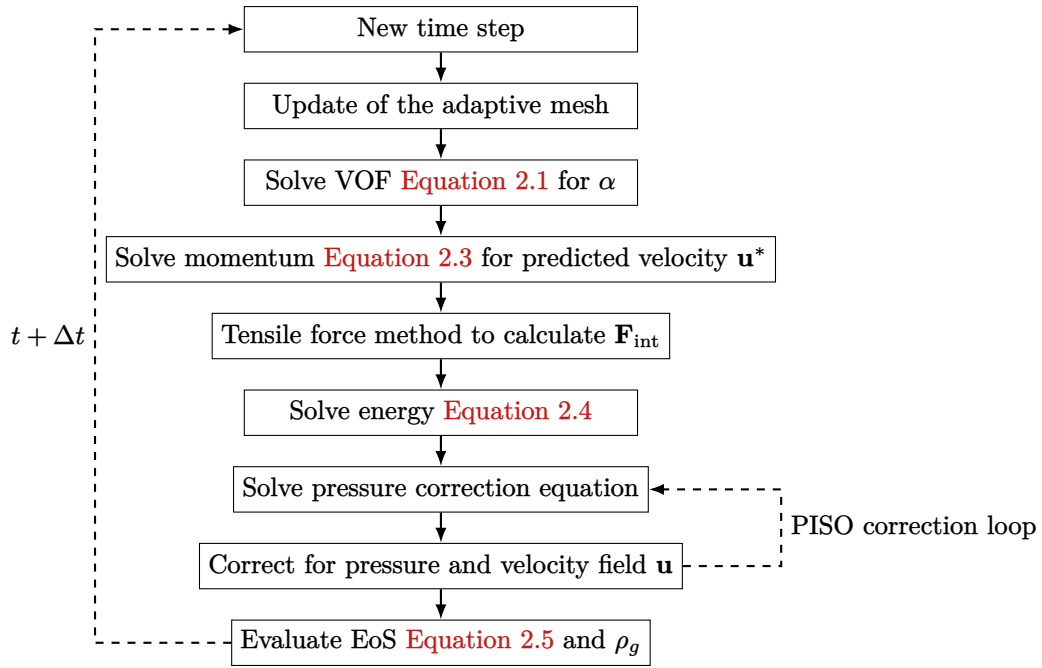


Figure 2.1: Simplified PISO solution procedure over a single time step, simplified from Thuy’s solution [6].

2.5 | Secondary Breakup and Coupling Approaches

With the relevant physical properties determined at each time-step, a complete simulation of the initial primary breakup may be solved using VOF. However, since primary and secondary breakup phases are modeled using distinct numerical approaches, this primary breakup data must be transferred to a separate secondary breakup simulation, which is not relevant to the findings of this report. However, the coupling approach of the primary breakup data with the secondary breakup simulation is necessary to understand.

To connect these models, a coupling plane is introduced. The coupling plane refers to a numerical boundary, rather than a distinct physical transition in the atomization process. At this location, the properties of droplets obtained from the primary breakup simulation are extracted and initialized in the secondary breakup simulation. To this end, there are two methods of coupling the primary and secondary breakup data: the Direct coupling approach (DCA), and the Statistical coupling approach (SCA).

In the direct coupling approach, individual droplets that fulfil the given extraction criteria are transferred directly to the secondary breakup simulation model. The relevant droplet properties: position, diameter, and velocity, are retained during this extraction. While this provides a high simulation accuracy over the full gas atomization process, it has high computational cost.

In the statistical coupling approach, which is considered in this report, individual droplets are not transferred directly. Rather, statistical distribution data of the relevant droplet properties, are obtained at a given transition plane. This statistical data is then transferred to be utilized in the secondary breakup simulation. This distinction is shown in Figure 2.2.

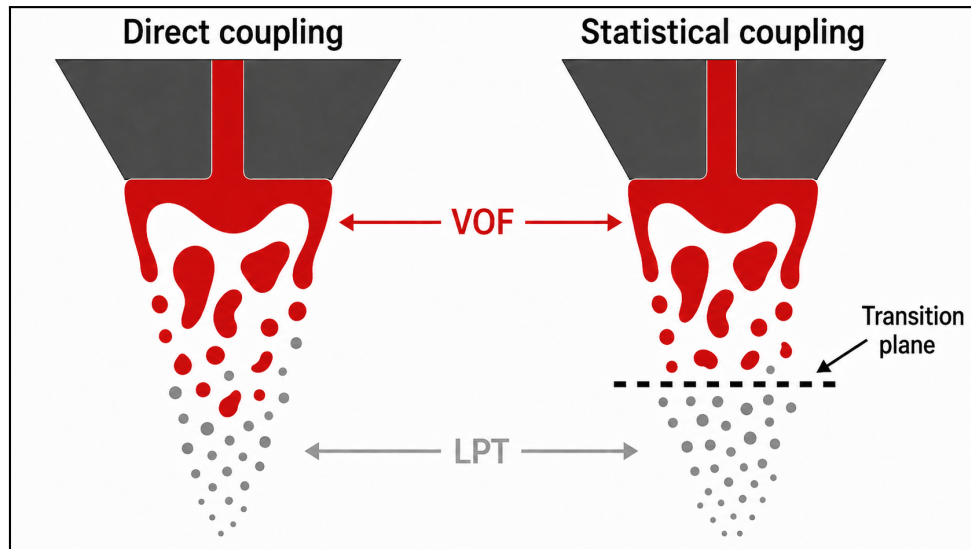


Figure 2.2: DCA and SCA visualized, adapted from Thuy [6]

The location of the transition plane must therefore be selected such that the droplets at this location are sufficiently small and spherical for the assumptions of the Lagrangian model utilized in the secondary breakup simulation to be valid. Primarily, this refers to the assumption that these droplets are discrete point particles.

3 | Simulation Setup

The simulation setup features two separate simulation setups and initial conditions. In general, the differences between the cases are primarily to do with the following variables:

- Nozzle geometry
- Liquid metal alloy
- Atomization gas
- Inlet conditions
- Simulation start-time and duration

3.1 | Case 1

For case 1, Nitrogen gas (N_2) is used as the atomization gas, while the liquid metal phase consists of molten aluminium. The geometry at the gas and melt inlets feature a close-coupled configuration, since the gas jet nozzle interacts with the falling liquid metal immediately after it leaves the nozzle. The gas nozzle is an annular slit, since the gas exits through a continuous annular opening around the liquid metal. This is shown below in [Figure 3.1](#).

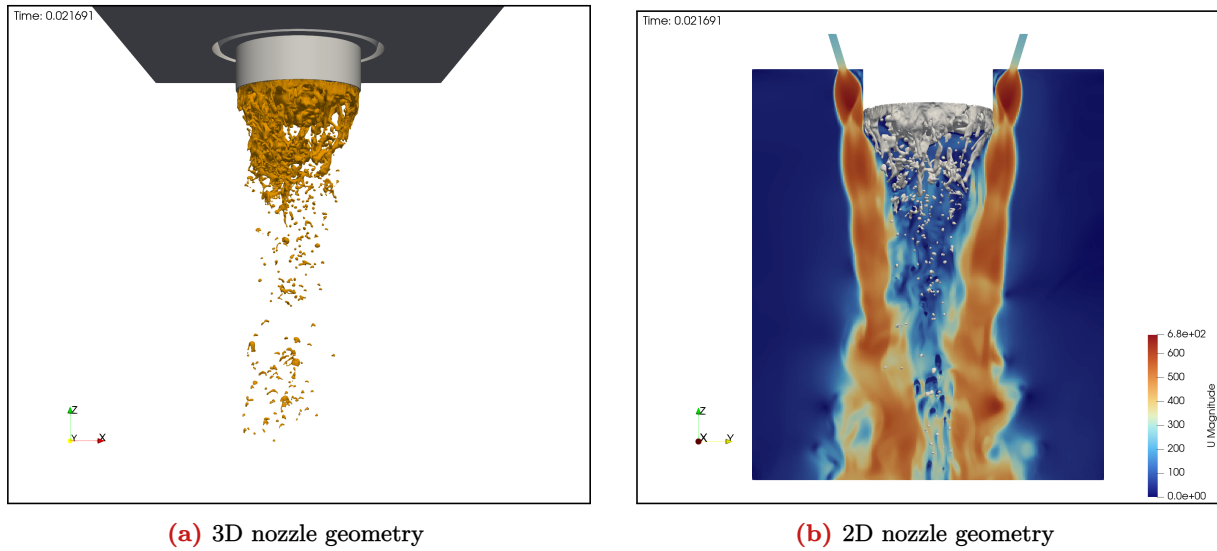


Figure 3.1: Nozzle geometry in Case 1

Furthermore, the simulation start time is ≈ 0.021 s into the simulation, hence the initial startup transient has mostly passed, and the flow starts from a statistically steady regime.

The material properties and initial simulation conditions are shown in [Table A.1](#) and [Table 3.1](#) respectively.

Table 3.1: Physical properties and initial conditions used in Case 1

Parameter	Value	Unit
Atomization gas pressure	10	bar
Atomization gas temperature	300	K
Melt flow rate	7.25×10^{-6}	m^3/s
Melt inlet temperature	1033	K
Melt superheat	100	K
Simulation start time	0.021250726	s
Simulated duration	2×10^{-3}	s

The relevant planes for axial distance relative to the nozzle are shown in **Figure 3.2**. It should be noted that the axial distance (d_z) values stated in the report correspond to the distance from the nozzle. Hence, $d_z = 0$ mm at the tip of the nozzle, and $d_z \approx 31$ mm at the bottom of the simulation.

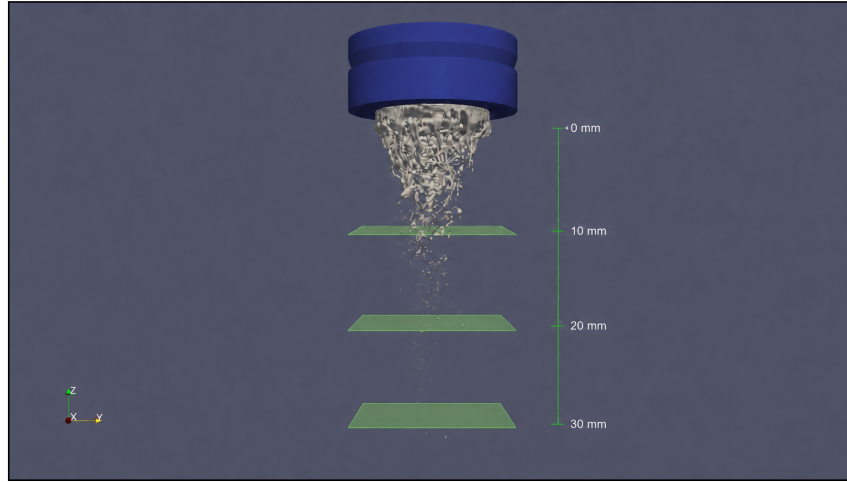


Figure 3.2: 10 mm, 20 mm, and 30 mm planes as shown in Case 1

3.2 | Case 2

For case 2, Argon gas (Ar) is used as the atomization gas, with the liquid metal phase as a Copper-Tin ($Cu - Sn$) melt. The nozzle geometry is distinct to that of case 1. The incoming gas is directed at 75 degrees. The nozzle geometry is shown in **Figure 3.3**.

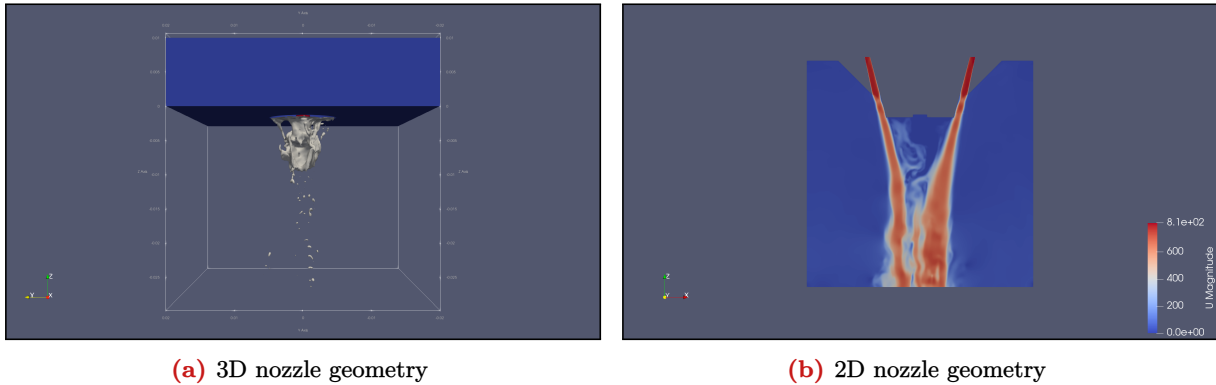


Figure 3.3: Nozzle geometry in Case 2

As shown, the gas jets are located closer to the melt nozzle, with a steeper slope into the melt nozzle.

Unlike case 1, the simulation start time is 0.01 s into the simulation, hence the initial startup transient is still present. The simulation takes a while to develop into a statistically steady regime.

The material properties and initial simulation conditions are shown in **Table A.2** and **Table 3.2** respectively.

Table 3.2: Physical properties and initial conditions used in Case 2

Parameter	Value	Unit
Atomization gas	Argon (Ar)	–
Liquid metal	Cu–Sn	–
Atmospheric pressure	1.0×10^5	Pa
Gas inlet total pressure	10.1	bar
Gas inlet temperature	302.15	K
Gas inlet angle	75	°
Melt inlet temperature	1609.15	K
Gas inlet boundary condition	Total pressure	–
Simulation start time	0.01	s
Simulated duration	3.39×10^{-3}	s

3.3 | Comparing Cases

Table 3.3: Comparison of the main material properties and inlet conditions for Case 1 and Case 2.

Parameter	Case 1	Case 2	Unit
<i>Gas phase</i>			
Atomization gas	N ₂	Ar	–
Molecular weight, M_g	28.01	39.95	g/mol
Heat capacity, $C_{p,g}$	979	520.33	J/(kg K)
Dynamic viscosity, μ_g	1.66×10^{-5}	2.2724×10^{-5}	Pa s
Prandtl number, Pr_g	0.67	0.664	–
Gas inlet pressure	10	10.1	bar
Gas inlet temperature	300	302.15	K
<i>Liquid phase</i>			
Liquid metal	Al	Cu–Sn	–
Reference density, $\rho_{0,l}$	2346.3	7720	kg/m ³
Heat capacity, $C_{p,l}$	921.1	556.29	J/(kg K)
Dynamic viscosity, μ_l	0.0011	0.004023	Pa s
Surface tension, σ	0.8545	1.09	N/m
Melting temperature, T_m	933	1332	K
Melt inlet temperature	1033	1609.15	K
Melt superheat	100	277.15	K
<i>Simulation Time</i>			
Simulation start time	0.021250726	0.01	s
Simulated duration	2×10^{-3}	3.39×10^{-3}	s
Number of data collection time-steps	101	413	–

The simulations for the two cases were performed separately, with OpenFOAM solving for flow-field data and liquid-structure information at each time-step. The data extraction procedure used to determine the results presented in [section 4](#) is explored further in the following [subsection 3.4](#).

The data utilized in this report is obtained from the VOF Cloud Data, which refers to the output data files generated by the software OpenFOAM through the runtime of the simulation. The numerical methodology that OpenFOAM utilizes to extract this data is detailed further below.

3.4 | Data Extraction

The simulation solver computes key properties within discrete cells over the full mesh. Each cell in the mesh contains the liquid volume fraction α as per [Equation 2.1](#). Within the simulation, liquid cells are identified using a minimum liquid volume fraction threshold of $\alpha_{threshold} = 1 \times 10^{-8}$, as imposed by Thuy [6]. For detection of liquid structures, cells that meet the volume-fraction threshold $\alpha > \alpha_{threshold}$ are

grouped collectively with surrounding connecting cells as a single VOF structure. These liquid structures may not be clearly defined as droplets yet - these may instead be ligaments, liquid sheets, satellite droplets, or any other irregularly shaped structures.

The volume of the given liquid structure V_j is then determined as the summation of the liquid volume fraction and cell volume (V_i) within the liquid structure as:

$$V_j = \sum_{i \in j} \alpha_i V_i \quad (3.1)$$

The simulation cell volume (V_i) is dependent on the local mesh refinement within the simulation, which is explored further by Thuy [6].

Using the liquid structure volume (V_j), the equivalent spherical droplet diameter (D) is determined as:

$$D = \left(\frac{6V_j}{\pi} \right)^{1/3} \quad (3.2)$$

This equivalent diameter (D) refers to the diameter of a given hypothetical sphere that has the equivalent volume of the given liquid structure [9].

The center of mass of the liquid structure is then determined:

$$\mathbf{x}_{\text{COM}} = \frac{\sum_i \alpha_i V_i \mathbf{x}_i}{\sum_i \alpha_i V_i} \quad (3.3)$$

where $\mathbf{x}_i$ is the position vector of the cell i . Following this, the minimum and maximum distances from the center of mass to the gas-melt interface boundary, $R_{\min}$ and $R_{\max}$ respectively, are determined.

Additionally, the droplet velocity $\mathbf{U}_{\text{drop}}$ is determined via the velocity field over the identified liquid volume, and the velocity of the gas surrounding the liquid structure $\mathbf{U}_{\text{cont}}$ is similarly determined. From this, the relative gas-droplet velocity U_{rel} is determined:

$$U_{\text{rel}} = |\mathbf{U}_{\text{cont}} - \mathbf{U}_{\text{drop}}| \quad (3.4)$$

Droplet temperature T is similarly determined through the temperature field associated with the given liquid structure.

The collected data is then written into the VOF Cloud Data for the given time-step, and then proceeds to the next time-step and repeats. The complete data extraction methodology is presented in Figure 3.4.

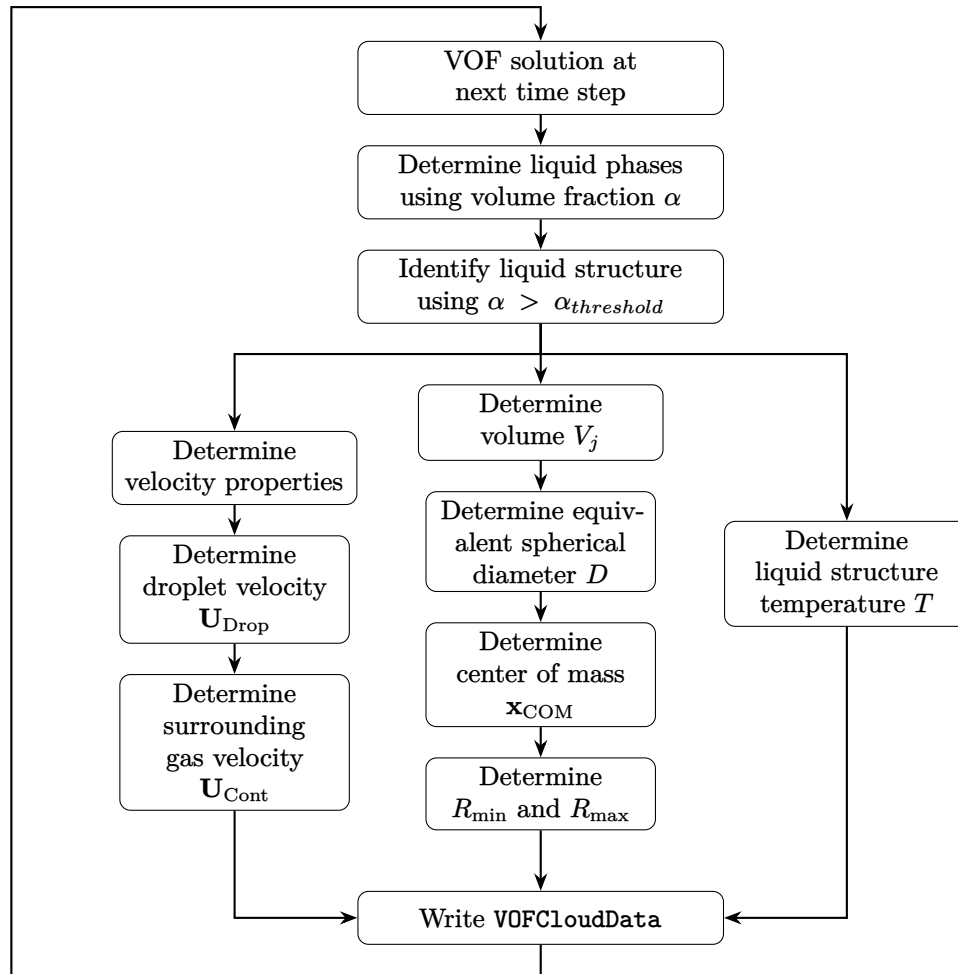


Figure 3.4: Flowchart of the methodology used to extract VOF cloud data

4 | Results

The raw simulation output was subsequently post-processed.

4.1 | Interpretation of Cases and Raw Data

The raw VOF cloud data contains properties of every detected liquid structure at each given time-step of the simulation. Due to the large quantity of liquid structures, this data exhibits considerable noise. Hence, the key properties extracted in both cases were grouped into distinct groups based on the axial distance relative to the nozzle.

The axial domain ranging from $0 \text{ mm} < d_z < 30 \text{ mm}$ was divided into distinct 1 mm bins. This axial domain is represented in [Figure 3.2](#). Each detected liquid structure was assigned to a bin according to the axial position of its center of mass relative to the nozzle (d_z). Every i th bin is defined as $b_i = [i, i + 1) \text{ mm}$, where $i = 0, 1, \dots, 30$.

The data within each 1 mm axial bin were then processed using the median and interquartile range (IQR). The median represents the 50th percentile of the data and was chosen since it is less sensitive to the wide distribution of values and outliers, which were present in the extracted datasets. The IQR, the difference between the 75th (Q_3) and 25th (Q_1) percentile of data, is expressed as $\text{IQR} = Q_3 - Q_1$. It provides a statistical basis for the error margin and represents the spread of the most relevant data.

Consequently, the interval between Q_1 and Q_3 contains the middle 50% of the data within each axial bin. In the presented figures, the median is shown as the central data point, while the error bars extend from the lower limit Q_1 , to the upper limit Q_3 .

Furthermore, the simulation time-steps utilized in the data collection were also considered. As stated in [Table 3.3](#), case 1 begins at time $t \approx 0.021 \text{ s}$ past the start of the simulation and contains 101 time-steps. Hence, the simulation is observed as being in a statistically developed state, i.e., it is past the point of the initial transient period. Hence, data from all 101 time-steps is utilized in the data.

Similarly stated in [Table 3.3](#), case 2 begins at time $t = 0.01 \text{ s}$ and contains 413 time-steps. There is some initial transient startup observed, both in the visual simulation and the data obtained. Hence, the time-steps considered are between $140 \leq t_{step} \leq 413$, as this dataset yields cleaner simulation output once the initial transient startup is past.

The following subsections detail the key properties extracted from the VOF cloud data, as well as the post-processing results and data interpretation.

4.2 | Droplet Diameter

The droplet diameter (D) is a critical parameter when assessing the suitability of a transition plane. As the droplets travel further downstream, i.e., increasing axial distance with respect to the nozzle, the liquid is hypothesized to gradually deform and break up from the liquid core due to the violent aerodynamic force transfer from the high-velocity gas. Hence, the droplet diameter is expected to rapidly decrease with downstream distance from the nozzle until a plateau is reached, beyond which the changes are minimal [\[10\]](#). At this point, the droplets are sufficiently stable to be transferred to and broken up in the secondary atomization phase [\[10\]](#).

The droplet diameter with respect to the axial distance in both cases is shown in [Figure 4.1](#).

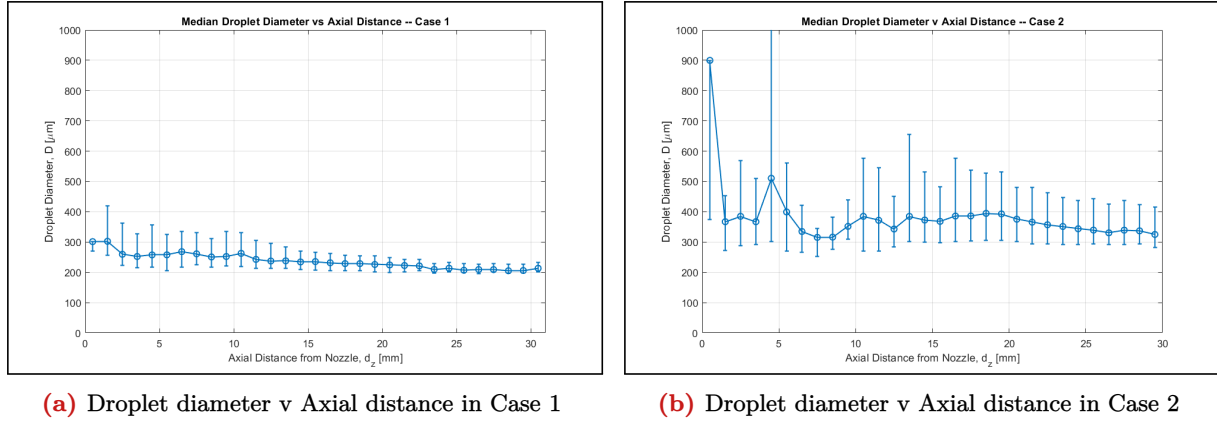


Figure 4.1: A decreasing droplet diameter trend is observed in Case 1. Case 2 is volatile initially but eventually follows a decreasing trend.

There is a slow and gradual decrease in droplet diameter as the axial distance from the nozzle increases. Case 1 features no major anomalies in the data - the droplet diameter follows a slow and gradual decrease and plateaus towards the end. Conversely, case 2 is noticeably more volatile, with large spikes at $d_z = 4$ mm and $d_z = 10$ mm. Past $d_z = 20$ mm, the data smooths out and follows this gradual decreasing trend.

The data obtained validates the stated hypothesis that droplet diameter gradually decreases and eventually plateaus.

4.3 | Droplet Velocity

Next, the droplet velocity (U_{drop}) is assessed. Initially, droplet velocity is hypothesized to increase with axial distance from the nozzle as the droplets are rapidly accelerated by momentum transfer from the surrounding high-velocity gas. As the droplets get further downstream, the rate of acceleration is expected to decrease as the droplet velocity approaches that of the gas phase, and the droplet velocity would be expected to gradually decay and decrease as the droplet velocity equalizes with the velocity of the surrounding gas (U_{cont}) [11].

The droplet velocity with respect to the axial distance in both cases is shown in Figure 4.2.

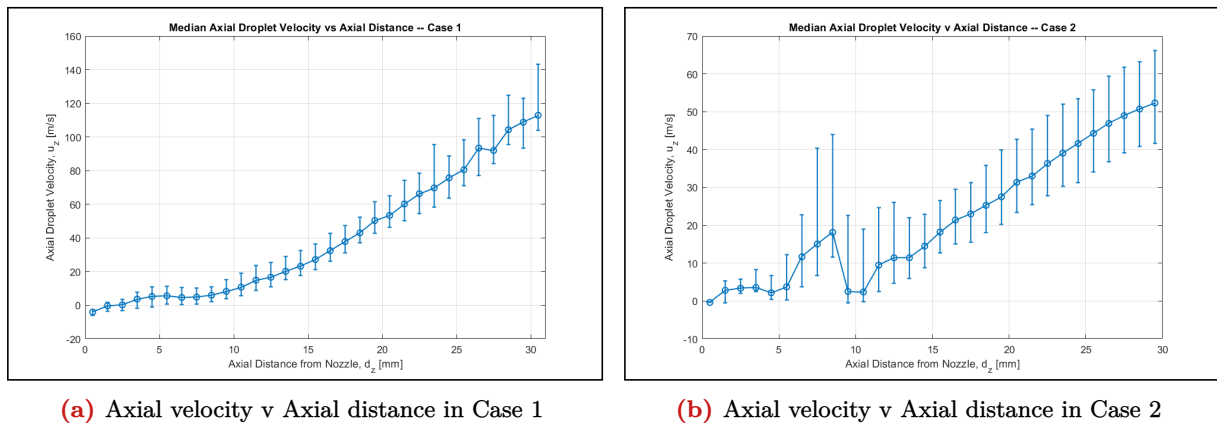


Figure 4.2: An increasing droplet axial velocity trend is observed in Case 1. Case 2 is initially volatile with a significant spike at $d_z = 8$ mm and crash afterwards. However, it eventually follows an increasing trend.

The droplet velocity in case 1 follows the increasing trend, however, the decreasing trend beyond a given point is not observed. In case 2, the droplet velocity is initially highly volatile, until $d_z \approx 15$ mm. There is a gradual increase from $d_z = 4$ mm until a large spike at $d_z = 8$ mm and a subsequent crash following it. This spike may be explained by potential backflow due to the high-velocity gas. Backflow, or

recirculation, refers to the rapid acceleration of liquid melt that occurs due to low-pressure regions formed by high-velocity gas close to the melt delivery nozzle [12]. This phenomenon is shown in Figure 4.3.

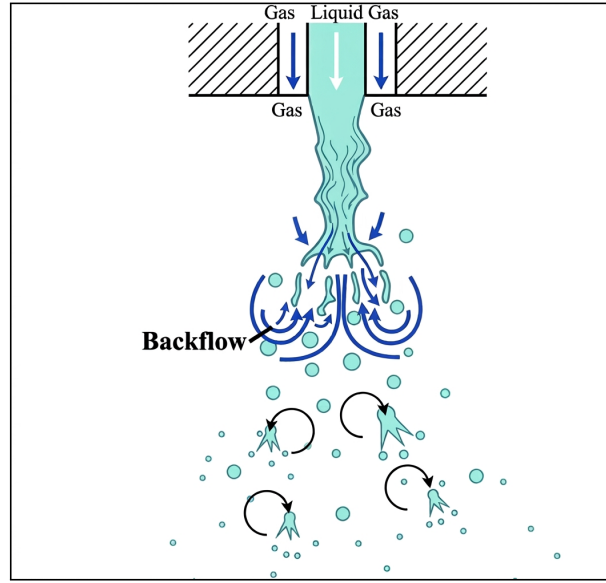


Figure 4.3: Backflow visualized. The image is AI-generated by ChatGPT [3], adapted from [13]

This recirculation may plausibly explain the sudden spike in droplet velocity. Once the droplets go past this recirculation zone, the droplet velocity returns to normal, which corresponds to $d_z = 9$ mm in Figure 4.2b.

4.4 | Droplet Temperature

Droplet temperature (T) is then explored. The droplet temperature is hypothesized to be highest at the nozzle, and gradually decrease downstream through the colder atomizing gas. The droplets are expected to equalize temperature with the cooler gas, as heat is removed from the droplet surface primarily through convection. This cooling behaviour is further dependent on droplet size (D) and the relative velocity u_{rel} - smaller droplets generally experience higher cooling rates due to the higher surface area to volume ratio [11].

The droplet temperature with respect to the axial distance in both cases is shown in Figure 4.4.

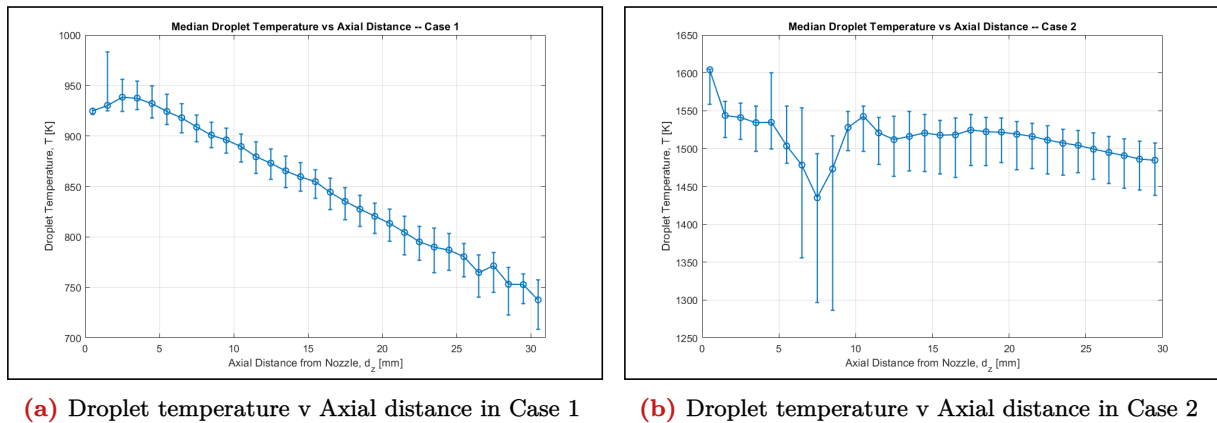


Figure 4.4: Decreasing droplet temperature trend observed in both Cases 1 and 2. Case 1 features a steep downward trend, whereas Case 2 dives from $d_z = 4$ mm and reaches its lowest value at $d_z = 8$ mm and subsequently increases until $d_z = 11$ mm, and then continues a gradual downward trend.

The droplet temperature is mostly stable in case 1. It increases slightly until $d_z \approx 3$ mm, and then decreases gradually. Case 2 is volatile, featuring a sudden dip in temperature at the same region where

droplet velocity spiked in [Figure 4.2b](#). It is likely that recirculation is an explanation for the sudden drop in temperature - whereby a jump in droplet velocity increases relative velocity (u_{rel}), which increases heat transfer from the droplets into the surrounding gas.

Observing the trends of properties thus far, there arises an anomaly within case 2 consistently over the same approximate domain $4 \text{ mm} \leq d_z \leq 10 \text{ mm}$, as shown in [Figure 4.1b](#), [Figure 4.2b](#), and [Figure 4.4b](#). This feature is observed throughout case 2 across several different variables, which makes it unlikely to be a coincidence. Recirculation is the most likely explanation.

4.5 | Sphericity

Sphericity is a measure of how closely a three-dimensional object resembles a perfect sphere. There are several expressions to define sphericity, the most common of which involves the ratio between the equivalent sphere surface area and the surface area of the object [14], however, since surface area data is not derived in the VOF cloud data, alternate definitions must be implemented.

In general, it is hypothesized that sphericity, regardless of mathematical expression implemented, increases as the axial distance from the nozzle increases. This indicates that droplets increasingly resemble perfectly spherical particles as they drop further down the nozzle.

There are various criteria to define sphericity. These criteria are discussed further below.

4.5.1 | Shape Criterion ψ

One such criterion is the shape criterion ψ . It is expressed as:

$$\psi = \frac{R_{max}}{D} \quad (4.1)$$

A lower value of ψ indicates a spherical droplet, and the perfect sphere corresponds to $\psi = 0.5$, whereas a higher value indicates an elongated or irregular droplet shape. The sphericity requirement $\psi < 1$ is applied to droplets to ensure that they are sufficiently spherical to be transferred from the transition plane, as done by Thuy [6].

The droplet shape criterion with respect to the axial distance in both cases is shown in [Figure 4.5](#).

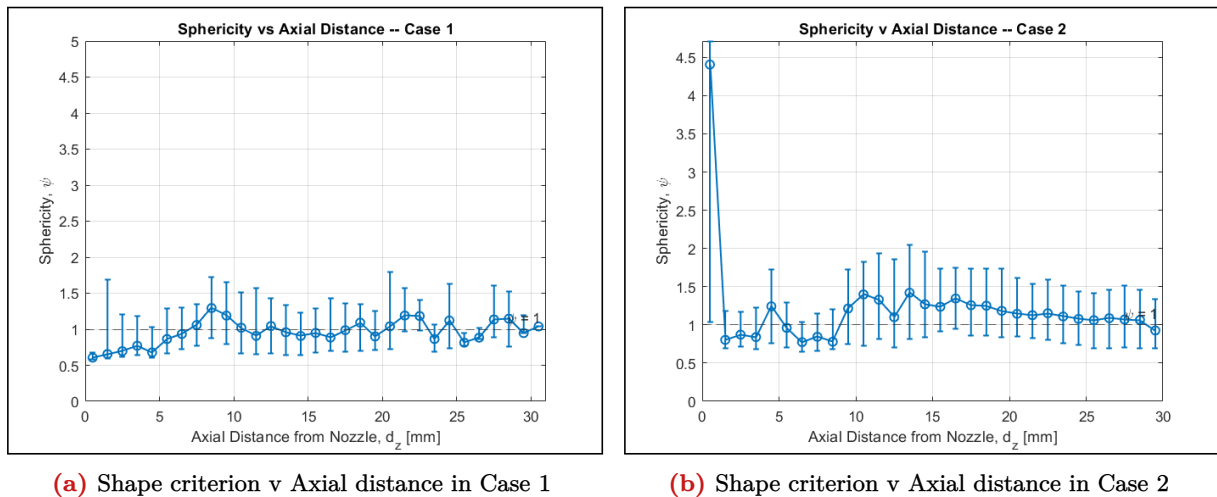


Figure 4.5: The droplet shape criterion trend is inconclusive. A volatile trend is observed in both Cases 1 and 2. However, both cases satisfy the shape criterion $\psi < 1$ in the end, at $d_z = 30 \text{ mm}$.

The shape criterion is inconsistent between the two cases. Case 1 indicates that particles are highly spherical close to the nozzle, and gradually become less spherical, hovering around the shape criterion $\psi < 1$ towards the latter half of the axial distance, as shown in [Figure 4.5a](#).

Case 2 features poor sphericity at $d_z = 1$ mm, however the $Q1$ value sits close to $\psi = 1$. Furthermore, $Q3 = 5$ is not significantly higher than the median, which indicates a large cluster of data between the median $\tilde{x} = 4.5$ and $Q3 = 5$. This is expected at the nozzle as larger droplets (see [Figure 4.1](#)) and more irregular liquid structures are formed as the liquid core begins the breakup process. The sphericity then begins to gradually decrease from $d_z \approx 20$ mm onward, and the shape criterion $\psi < 1$ is fulfilled at $d_z = 29$ mm.

This dataset is inconclusive to make a claim regarding sphericity and axial distance. Hence, another criterion for sphericity, the aspect ratio, is considered.

4.5.2 | Aspect Ratio

The aspect ratio is another expression of sphericity, as it provides a measure of the geometric deformation of the liquid structures. It is defined as:

$$AR = \frac{R_{\min}}{R_{\max}} \quad (4.2)$$

For a perfectly spherical droplet, $R_{\min} = R_{\max}$ and therefore $AR = 1$. Lower values indicate increasing deviation from a spherical shape with $AR = 0$ indicating poor sphericity.

The droplet aspect ratio with respect to the axial distance in both cases is shown in [Figure 4.6](#).

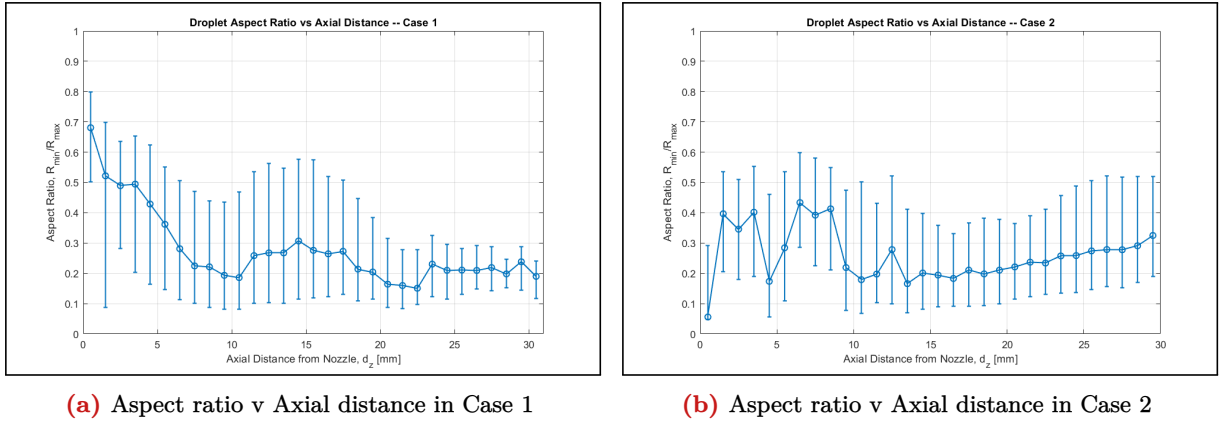


Figure 4.6: There is no clear aspect ratio trend in either Case 1 or Case 2. Both cases are volatile. Case 1 does seem to trend downward which indicates poor sphericity, whereas Case 2 seems to trend upward which indicates higher sphericity.

This data is similarly inconclusive. Both cases are fairly volatile and no clear observations or conclusions may be drawn from these results. Case 1 shows that at the furthest axial distance from the nozzle, aspect ratio $AR \approx 0.2$, which is poor sphericity, as shown in [Figure 4.6a](#). Conversely, case 2 highlights an increasing trend of aspect ratio towards the furthest axial distance, however, the aspect ratio $AR \approx 0.3$ is still quite low.

No clear conclusion on sphericity may be drawn from the aspect ratio. Hence, the shape deviation is considered next.

4.5.3 | Shape Deviation ψ_{dev}

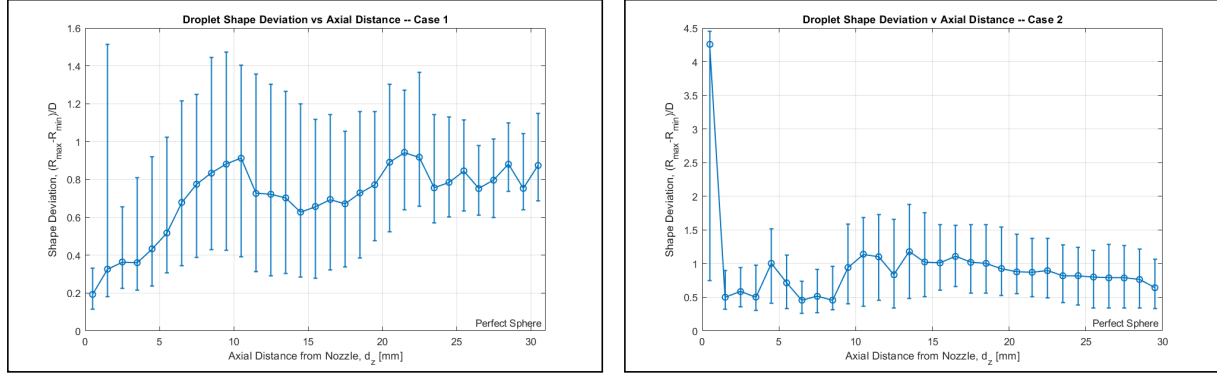
The shape deviation provides an additional measure of how far an identified liquid structure deviates from a spherical geometry. It is defined as:

$$\psi_{\text{dev}} = \frac{R_{\max} - R_{\min}}{D} \quad (4.3)$$

For a perfectly spherical droplet, $R_{\min} = R_{\max}$ and therefore $\psi_{\text{dev}} = 0$. Increasing values of ψ_{dev} indicate greater geometric deformation and poorer sphericity.

It is hypothesized that the shape deviation decreases with increasing axial distance from the nozzle. This corresponds to the large irregular liquid structures close to the nozzle gradually breaking up into smaller, more spherical droplets. Consequently, ψ_{dev} is expected to approach zero with increasing axial distance.

The droplet shape deviation with respect to the axial distance in both cases is shown in Figure 4.7.



(a) Shape deviation v Axial distance in Case 1

(b) Shape deviation v Axial distance in Case 2

Figure 4.7: Inconclusive shape deviation trend observed in both Cases 1 and 2. Case 1 trends upwards with axial distance, whereas Case 2 trends downwards.

There is no clear conclusion that may be drawn from the shape deviation results. The next, and final, sphericity criterion is considered in radial convergence.

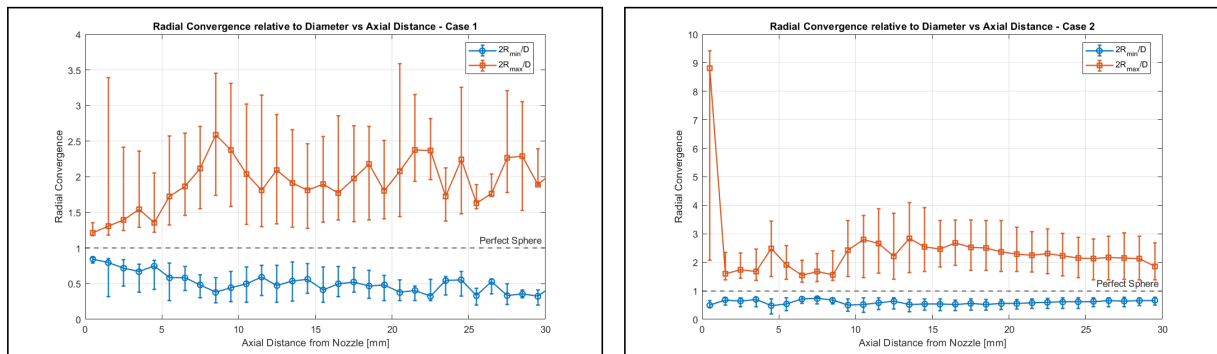
4.5.4 | Radial Convergence

The radial convergence criterion provides a measure of the extent to which the minimum and maximum radii of a liquid structure approach an equivalent sphere. The normalized radial extents are defined as

$$\frac{2R_{min}}{D} \quad \text{and} \quad \frac{2R_{max}}{D} \quad (4.4)$$

For a perfectly spherical droplet, $R_{min} = R_{max} = D/2$, such that both values converge at $\frac{2R_{min}}{D} = \frac{2R_{max}}{D} = 1$. Convergence of the two values towards 1 indicates higher sphericity over the axial distance.

The radial convergence with respect to the axial distance in both cases is shown in Figure 4.8.



(a) Radial convergence v Axial distance in Case 1

(b) Radial convergence v Axial distance in Case 2

Figure 4.8: Case 1 diverges, which indicates poor sphericity over the axial distance. Case 2 shows the expected convergence, which indicates high sphericity over the axial distance.

Case 1 shows a divergence in the normalized radial extents, which indicates poor sphericity at distances further from the nozzle. In particular, $\frac{2R_{max}}{D}$ is particularly volatile; therefore, no clear trend for it may be ascertained. Case 2 shows convergence towards further axial distances, which is the hypothesized trend.

However, it is not possible to draw a conclusion on sphericity from this data either.

To conclude with sphericity, there is no clear relationship between droplet sphericity and axial distance from the nozzle that may be established from the findings presented.

4.6 | Weber Number

The Weber number (We) is a key parameter in determining spray and droplet behavior. It is a reference measure to observe the ratio between the aerodynamic forces acting on a liquid structure and its surface-tension forces [15]. The Weber number is determined as follows:

$$We = \frac{\rho_g U_{rel}^2 D}{\sigma} \quad (4.5)$$

where ρ_g is the gas density, and σ is the surface tension. Larger Weber numbers indicate that the aerodynamic forces are increasingly dominant relative to surface tension and therefore suggest a greater tendency for droplet deformation and breakup. Conversely, smaller Weber numbers indicate that the surface-tension forces within the droplet dominate and that the droplet is less likely to undergo deformation and breakup.

It is hypothesized that the Weber number decreases with increasing axial distance from the nozzle. Close to the nozzle, a high relative velocity between the high-speed atomizing gas and liquid melt is expected. This corresponds to a high Weber number at this point since the Weber number is proportional to the relative velocity squared $We \propto U_{rel}^2$. Further downstream, the droplets accelerate and droplet velocity increases, and gas velocity gradually decreases, which causes the relative velocity to decrease. This results in a lower Weber number due to the same proportionality. Hence, it is expected that the Weber number decreases with axial distance.

The Weber number with respect to the axial distance in both cases is shown in Figure 4.9.

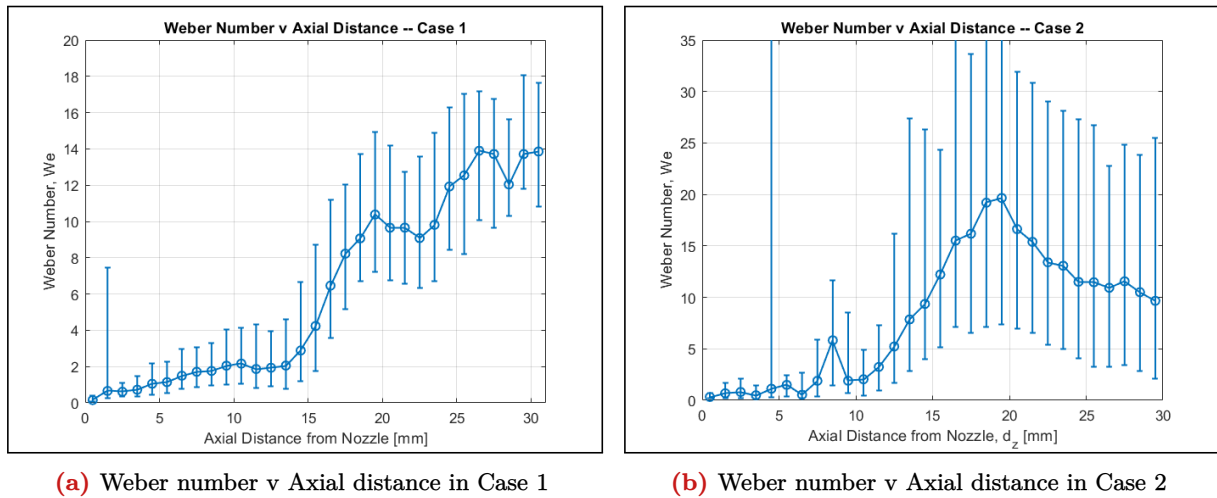


Figure 4.9: Inconclusive Weber number trend observed between Cases 1 and 2. The Weber number in Case 1 is up-and-down but seems to follow an increasing trend overall. Case 2 features a gradual increase until $d_z = 20$ mm, and then a gradual decrease at further axial distances.

The Weber number trend is similarly inconclusive between both cases. There is an inconsistent upwards trend in case 1, with dips at $d_z = 23$ mm and $d_z = 28$ mm. However, in case 2, the Weber number gradually increases until $d_z = 20$ mm, at which point it peaks and gradually decreases from then onward. Neither case matches the hypothesized decreasing relationship between We and axial distance.

It must be noted that the calculated Weber number was derived using approximate values for ρ_{gas} and surface tension σ . These values were solved in OpenFOAM using constant values, rather than localized variables that change dynamically with the simulation. The derived Weber number from the simulation is

therefore an approximation of the actual Weber number, which may have altered the findings determined from this data.

4.7 | Breakup Length

The breakup length (L_b) refers to the axial distance from the nozzle to the point where the continuous liquid metal core fully disintegrates into ligaments and droplets. The extent of the primary breakup region is usually characterized by the breakup length [16].

In this report, the breakup length was determined using a Python script that visually determines the axial distance from the nozzle to the tip of the liquid core at a given plane over each time-step. One such screenshot is shown in [Figure 4.10](#). This script was developed by [17].



Figure 4.10: The breakup length determination process - the axial distance (d_z) between the nozzle and the lowest point of the red outline is determined as the breakup length for the given time-step. This process is repeated for each time-step in the simulation.

The breakup length with respect to the simulation time in Case 2 is shown in [Figure 4.11](#).

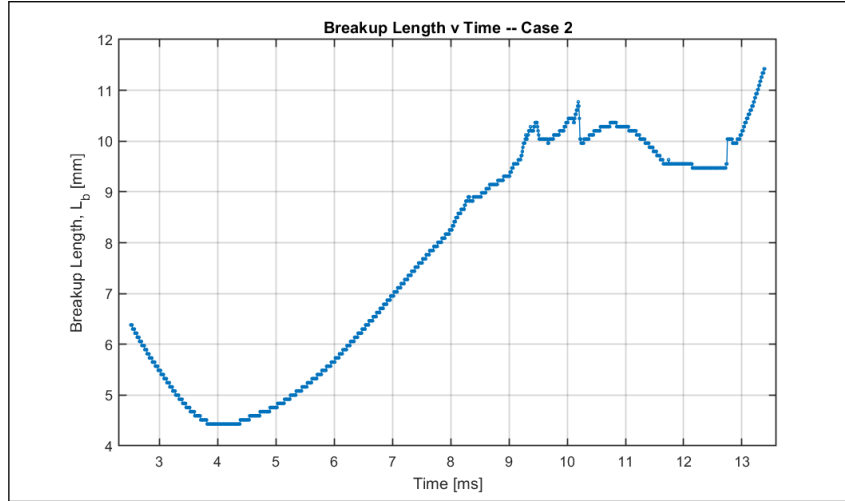


Figure 4.11: Breakup length v Simulation time for Case 2

The breakup length was only collected for Case 2. Furthermore, it should be noted that this data is available over a larger simulation duration compared to the results obtained for the other simulated values. Simulation data exists in the duration between 2.51 ms until 13.39 ms.

The breakup length initially features a sharp drop initially between 2 – 4 ms, and then gradually rises between 4 – 10 ms, before briefly settling at around 10 mm, and then jumping upwards towards the end of the simulation time. The sharp drop early may be that the atomization process is still undergoing the startup transient phase during this time and is not representative of the later regime. As the gas and melt flow develop, the liquid core gradually extends farther downstream before breaking up, however it does not seem to settle into a steady state within this simulation duration.

This data provides a lower bound for the placement of the transition plane. Since the transition plane cannot intersect the liquid metal core, the transition plane must be positioned downstream of the breakup length. From the data derived in this simulation, the maximum breakup length of the simulation at $L_b \approx 12\text{mm}$ defines the minimum axial distance at which a transition plane may be considered.

4.8 | Transition Plane

From the findings of the above sections, there is evidence to support a general guideline for the location of the transition plane. An effective transition plane must meet the criteria of resolving droplet breakup, i.e., it is unlikely for the droplet to break up any further in a primary breakup simulation utilizing VOF. Furthermore, that the transition plane is placed such that the droplet diameter and sphericity are close enough to reliably approximate these droplets as small particles to be transferred to the secondary breakup simulation using LPT.

To this end, several findings have been made to assert a claim on a transition plane. These are as follows:

- Through the findings from [subsection 4.7](#), a minimum distance for the transition plane is stated at an axial distance $d_z = 12\text{ mm}$ from the nozzle, as seen in [Figure 4.11](#). This states that the transition plane must be placed at an axial distance $d_z > 12\text{ mm}$ from the nozzle.
- The findings from [subsection 4.2](#) indicate that droplet diameter decreases gradually as the axial distance from the nozzle increases. This decreasing trend is observed from $d_z = 20\text{ mm}$ onward.
- The findings from [subsection 4.3](#) indicate an increasing trend of droplet axial velocity relative to axial distance. This trend is also shown in both cases from $d_z = 15\text{ mm}$ onward.
- The findings from [subsection 4.4](#) indicate a decreasing trend of droplet temperature relative to axial distance. This trend is shown in both cases from $d_z = 15\text{ mm}$ onward.
- The findings from [subsection 4.5](#) did not yield strong claims on the relationship between sphericity and axial distance, although the shape criterion requirement $\psi < 1$ was met for both cases at the

furthest axial distance $d_z = 30$ mm. However, this cannot be considered in the discussion for the transition plane since additional data points would be necessary to make a conclusive claim.

From these findings, a transition plane located **20 mm** downstream from the nozzle is recommended. This recommendation is driven primarily by the findings from the droplet diameter (D), axial velocity (u_z), and droplet temperature (T), and breakup length (L_b). However, it must be noted that this value is simply for these cases, it is certainly not a value that can be prescribed to any case of gas atomization. This depends heavily on the material properties, initial conditions, nozzle geometry, and numerous other factors.

A more general recommendation for the transition plane would be to observe the point at which:

- droplet diameter trends downwards and plateaus,
- axial velocity increases and then gradually decreases - this point was not reached in either case,
- and droplet temperature trends downwards

The earliest point at which all of these trends are observed would be the ideal transition plane of a given simulation.

5 | Conclusion

To conclude, the relevance of gas atomization has been established through its application in a broader manufacturing context, in particular MAM. The numerical and computational model of the primary breakup phase of gas atomization has been explored. Two simulation cases, with unique material properties, initial conditions, nozzle geometries, and simulation times, have been investigated. Through this, key properties of the gas atomization were explored further, primarily the droplet diameter (D), axial velocity (u_z), and droplet temperature (T), and the breakup length (L_b).

From these findings, clear trends in the above parameters are substantiated. The findings for sphericity (ψ , AR , ψ_{dev} , and radial convergence) and breakup length (L_b) require additional data points to make a confident assessment. The Weber number (We) findings proved inconclusive, however this is plausibly because the approximate values of gas density ρ_g and surface tension σ were implemented in the calculation. It is likely that a clearer trend would be observed if the local values were instead used to determine the Weber number.

In any case, these findings were utilized to provide a recommendation for the transition plane at an axial distance 20 mm downstream from the nozzle. However, this recommendation is highly specific to these cases. A broader recommendation was highlighted to determine a transition plane by observing the earliest point at which the droplet diameter and temperature trend downwards, and the axial velocity (u_z) trends upwards, and the breakup length utilized as the lower bound value. This optimizes the simulation model for reliability and computational efficiency.

To answer the research questions:

RQ1. How do droplet properties evolve relative to axial distance?

A1. As axial distance from the nozzle increases: droplet diameter decreases, droplet axial velocity increases, and droplet temperature decreases. The trends pertaining to droplet sphericity and Weber number are inconclusive from the results obtained in this study.

RQ2. How may trends in droplet properties relative to axial distance be used to determine an optimal transition plane?

A2. Trends in key droplet properties must first be observed. The earliest possible axial position relative to the nozzle where the desired properties are observed relates to the optimal location of the transition plane.

RQ3. Which additional parameters may be extracted to support the location of the transition plane?

A3. The breakup length has been utilized to determine a minimum location of the transition plane.

5.1 | Recommendation and Follow-Up

The conclusions drawn for the various properties would be greatly strengthened through additional data points, in particular by increasing the simulation workspace to extend further along the axial (z) axis. This is especially noticeable in [Figure 4.9](#) and [Figure 4.11](#), where additional axial data points would likely clarify these trends with more detail.

Furthermore, the sphericity methodology is not fully indicative of geometric sphericity. The data processing involves 2-D values, namely R_{min} , R_{max} , and D . A more relevant methodology would be to determine sphericity as the ratio between the equivalent spherical surface area of the particle and the actual surface area of the particle, as proposed [\[14\]](#). This is shown in [Equation 5.1](#):

$$\Psi_{ideal} = \frac{\text{Surface area of equal-volume sphere } (A_s)}{\text{Actual surface area of the particle } (A_p)} = \frac{\pi^{1/3}(6V_p)^{2/3}}{A_p} \quad (5.1)$$

Where $\psi_{ideal} = 1$ indicates a perfect sphere, and $\psi_{ideal} = 0$ tends towards worse sphericity. However, the particle surface-area data was not collected in the VOF cloud data, therefore this relationship could

not be used. Hence, a complete assessment of sphericity could not be made. It is possible that a clearer conclusion for sphericity could be drawn from utilizing the surface area data of the droplets.

Furthermore, additional cases involving separate metal melts and gases may be considered to provide further substantiated evidence for the selection process of a transition plane.

6 | References

- [1] Jasgurpreet Singh Chohan Rakesh Kumar, Manoj Kumar. The role of additive manufacturing for biomedical applications: A critical review. <https://www.sciencedirect.com/topics/engineering/additive-manufacturing>, n.d.
- [2] Linde AMT. What are metal powders? <https://www.linde-amt.com/resource-library/articles/metal-powders/>, n.d. Accessed: 10 August 2026.
- [3] OpenAI. Chatgpt, 2026. AI-generated image.
- [4] Mikhael Gorokhovski and Marcus Herrmann. Modeling primary atomization. *Annual Review of Fluid Mechanics*, 40:343–366, 01 2008.
- [5] Wentao Liu, Jiashun Liang, Shizhen Chen, Hui Zhao, and Haifeng Liu. Secondary atomization characteristics of droplets in strong shear airflow. *International Journal of Mechanical Sciences*, page 112013, 2026.
- [6] Dennis Pieter Leonard Thuy. *Numerical metallurgy: a modeling framework for gas atomization of molten metal*. PhD thesis, Eindhoven University of Technology, 2026.
- [7] Junji Shinjo. Recent advances in computational modeling of primary atomization of liquid fuel sprays. <https://www.mdpi.com/1996-1073/11/11/2971>, n.d. Accessed: 10 August 2026.
- [8] Science Notes. Surface tension – definition, examples, formula. <https://sciencenotes.org/surface-tension-definition-examples-formula/>, 2026. Accessed: 10 August 2026.
- [9] Anne Virden. A guide to equivalent spherical diameter for particle sizing. <https://www.malvernpanalytical.com/en/learn/knowledge-center/insights/a-guide-to-equivalent-spherical-diameter-for-particle-sizing>, July 2026. Malvern Panalytical.
- [10] A. Kastengren, J. Ilavsky, Juan Pablo Viera, Raul Payri, D.J. Duke, A. Swantek, F. Zak Tilocco, N. Sovis, and C.F. Powell. Measurements of droplet size in shear-driven atomization using ultra-small angle x-ray scattering. *International Journal of Multiphase Flow*, 92:131–139, 2017.
- [11] Baolong Zheng, Yaojun Lin, Yizhang Zhou, and Enrique J. Lavernia. Gas atomization of amorphous aluminum: Part i. thermal behavior calculations. *Metallurgical and Materials Transactions B*, 40(5):768–778, 2009.
- [12] Jiaqi Liu, Pu Wang, Yannan Dong, Huan Zhao, Jing Pang, and Jiaquan Zhang. Investigation on close-coupled gas atomization for fe-based amorphous powder production via simulation and industrial trials: Part i. melt breakup behaviors during primary atomization. *Journal of Materials Research and Technology*, 27:6568–6580, 2023.
- [13] Daehyun Choi, Jungwon Byun, and Hyungmin Park. Analysis of liquid column atomization by annular dual-nozzle gas jet flow. *Journal of Fluid Mechanics*, 943, 06 2022.
- [14] D. Wang and L.-S. Fan. 2 - particle characterization and behavior relevant to fluidized bed combustion and gasification systems. In Fabrizio Scala, editor, *Fluidized Bed Technologies for Near-Zero Emission Combustion and Gasification*, Woodhead Publishing Series in Energy, pages 42–76. Woodhead Publishing, 2013.
- [15] J. Shinjo. Recent advances in computational modeling of primary atomization of liquid fuel sprays. *Energies*, 11(11):2971, 2018.
- [16] T. Cheng, R. Leibovici, B. Kong, and R. van Hout. Experimental investigation of primary breakup in close-coupled gas atomization. *International Journal of Multiphase Flow*, 181:105009, 2024.
- [17] Stefan van Laarhoven. Analyze planes, python script, 2026. Unpublished code, personal communication.

A | Appendix A: Material Properties of Cases

Table A.1: Material properties used in the simulation.

Property	Symbol	Value	Unit
<i>Nitrogen gas</i>			
Molecular weight	M_g	28.01	kg/kmol
Heat capacity	$C_{p,g}$	979	J/(kg K)
Dynamic viscosity	μ_g	1.66×10^{-5}	Pa s
Prandtl number	Pr_g	0.67	–
<i>Aluminium melt</i>			
Molecular weight	M_l	28.9	kg/kmol
Reference density	$\rho_{0,l}$	2346.3	kg/m ³
Heat capacity	$C_{p,l}$	921.1	J/(kg K)
Heat of fusion	H_f	3.96×10^5	J/kg
Dynamic viscosity	μ_l	0.0011	Pa s
Prandtl number	Pr_l	0.0043	–
Surface tension	σ	0.8545	N/m
Melting temperature	T_m	933	K

Table A.2: Material properties used in Case 2.

Property	Symbol	Value	Unit
<i>Argon gas</i>			
Molecular weight	M_g	39.95	g/mol
Heat capacity	$C_{p,g}$	520.33	J/(kg K)
Heat of formation	$H_{f,g}$	0	J/kg
Dynamic viscosity	μ_g	2.2724×10^{-5}	Pa s
Thermal conductivity	k_g	0.017804	W/(m K)
Prandtl number	Pr_g	0.664	–
<i>Cu–Sn melt</i>			
Molecular weight	M_l	65.37	g/mol
Speed of sound	c_l	3140.17	m/s
Fluid constant	R_l	6127.87	J/(kg K)
Reference density	$\rho_{0,l}$	7720	kg/m ³
Heat capacity	$C_{p,l}$	556.29	J/(kg K)
Heat of formation	$H_{f,l}$	0	J/kg
Thermal conductivity	k_l	84.5	W/(m K)
Dynamic viscosity	μ_l	0.004023	Pa s
Prandtl number	Pr_l	0.0265	–
Surface tension	σ	1.09	N/m
Melting temperature	T_m	1332	K